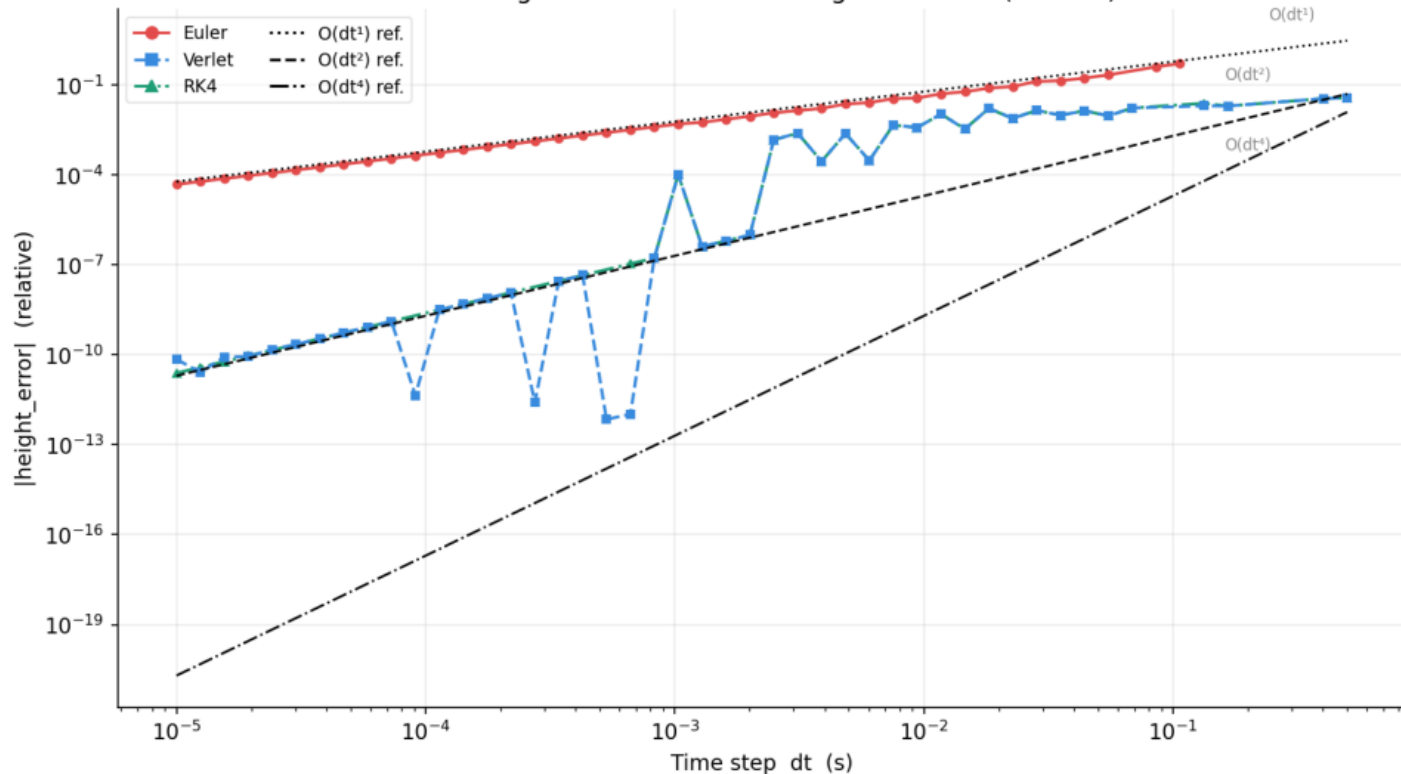


Peak height error vs dt — convergence order (e = 1.0)



CONVERGENCE ORDER (e = 1.0)

Measured log-log slopes:

Euler $\rightarrow O(dt^{0.98})$ $R^2=1.000$

Verlet $\rightarrow O(dt^{2.50})$ $R^2=0.795$

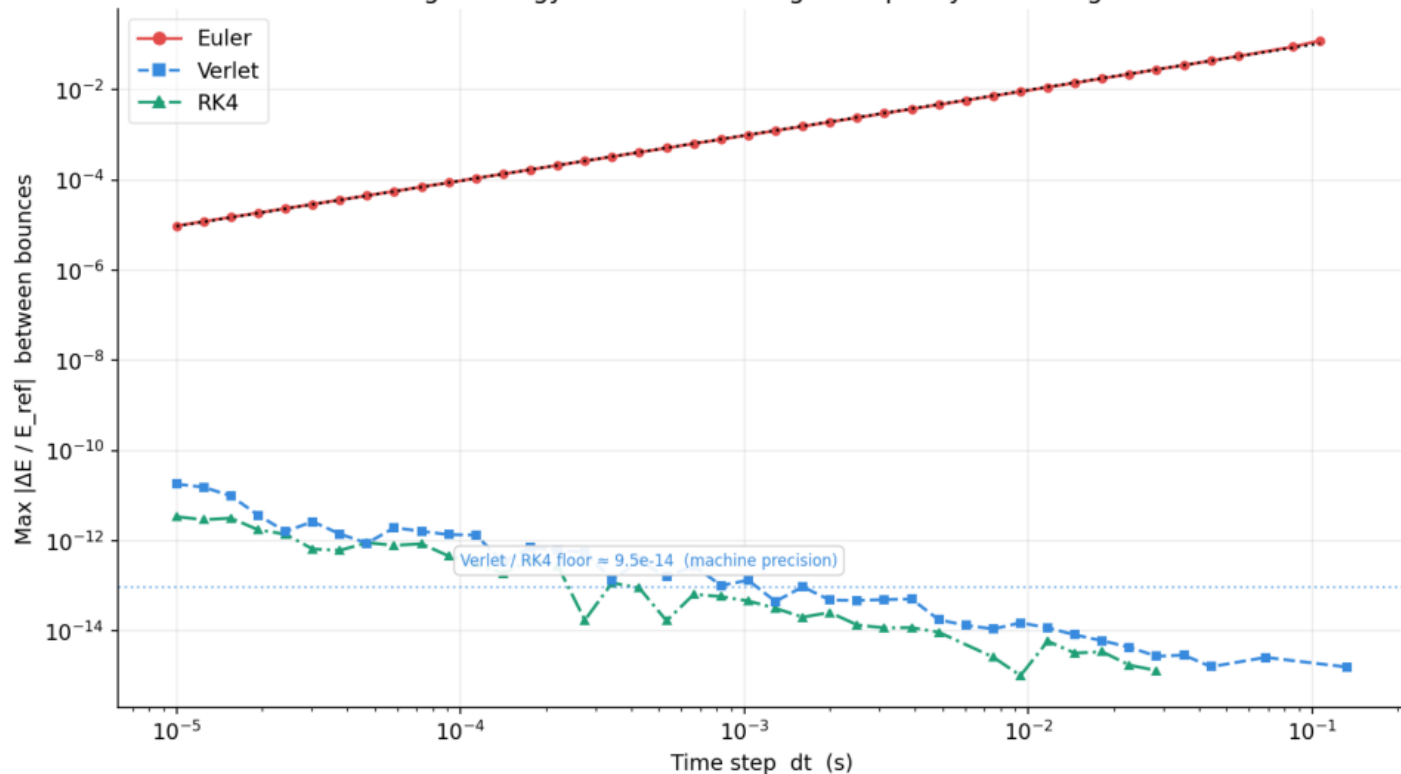
RK4 $\rightarrow O(dt^{2.38})$ $R^2=0.918$

With $e = 1.0$, the $O(dt)$ contact-timing error cancels for exact-position integrators (Verlet/RK4) $\rightarrow$ residual is $O(dt^2)$.

Euler: $O(dt)$ position error breaks the cancellation $\rightarrow O(dt^1)$ peak-height error.

RK4 does NOT show $O(dt^4)$: both Verlet and RK4 are exact for quadratic trajectories; the $O(dt^2)$ residual comes from the contact geometry, not the ODE integrator order.

Flight energy drift vs dt — integrator quality in free flight



FLIGHT ENERGY DRIFT

Integrator error in free flight.
Physical bounce losses excluded
(reference reset after each bounce).

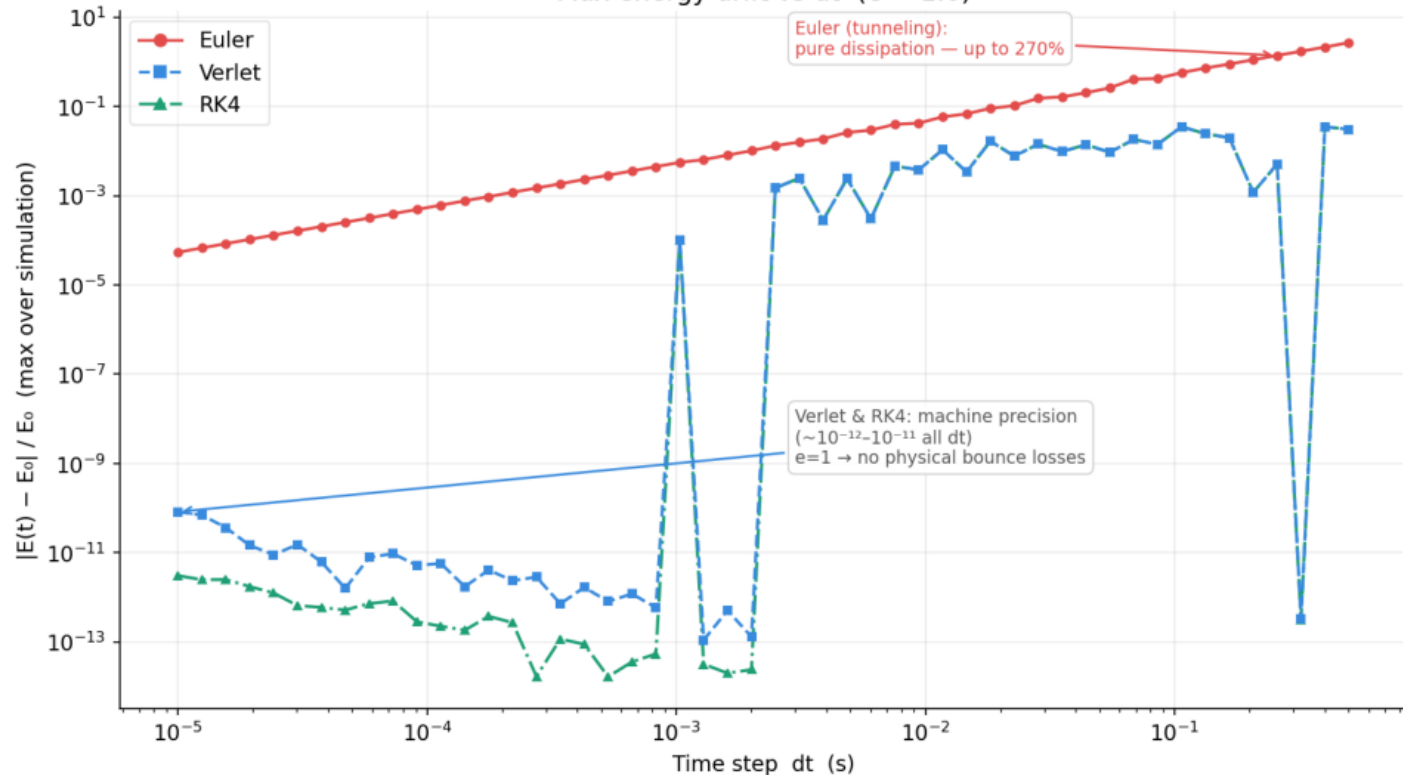
For constant-gravity free fall
(quadratic trajectory):

Euler $\rightarrow 0(dt)$ drift
slope = 1.01 $R^2 = 1.000$
 $-\frac{1}{2}g^2dt^2$ dissipation per step.

Verlet $\rightarrow$ exact (machine ϵ)
RK4 $\rightarrow$ exact (machine ϵ)

Conclusion: Verlet is optimal –
same flight accuracy as RK4
at $\frac{1}{4}$ the per-step FLOP cost.

Max energy drift vs dt (e = 1.0)



ENERGY CONSERVATION (e = 1.0)

With e = 1, there is NO physical energy loss at bounces.
All drift is NUMERICAL.

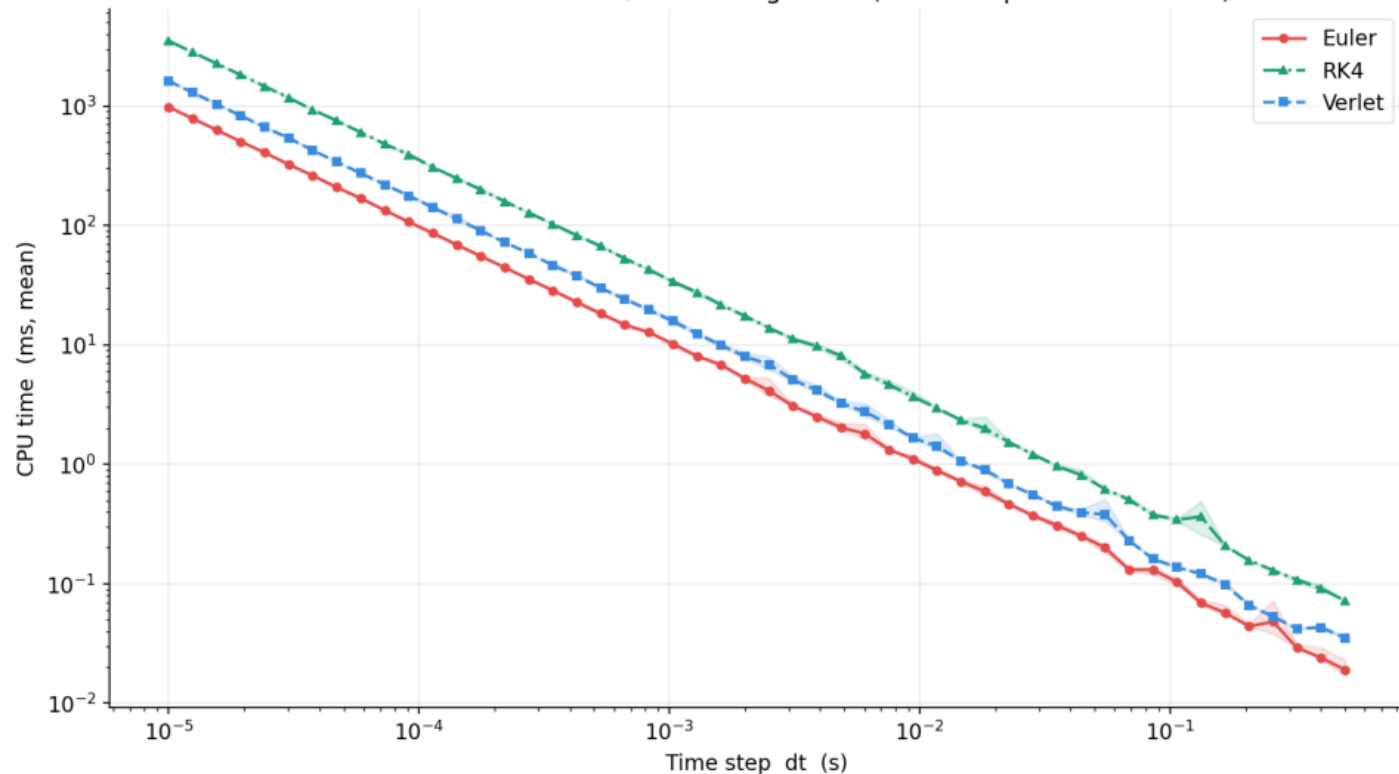
Verlet & RK4 (bouncing regime):
Machine precision $\sim 10^{-12}$.
Exact free-flight + elastic bounce → near-zero error.

Euler (bouncing regime):
O(dt) drift from $-\frac{1}{2}g^2dt^2/\text{step}$.
No plateau: e=1 removes the masking from physical loss.

Large dt – tunneling:
Euler: drift up to 270%.
Verlet/RK4: near zero.

→ Energy gap: $\sim 10^8\times$ at fine dt.

CPU time vs dt — with min/max timing band (1 warm-up + 3 timed runs)



CPU TIME (mean $\pm$ min/max band)

All solvers scale as $O(1/dt)$:
halving dt doubles CPU time.

Cost ratios vs Euler at median dt :

Verlet : $\sim 1.66\times$

RK4 : $\sim 3.35\times$

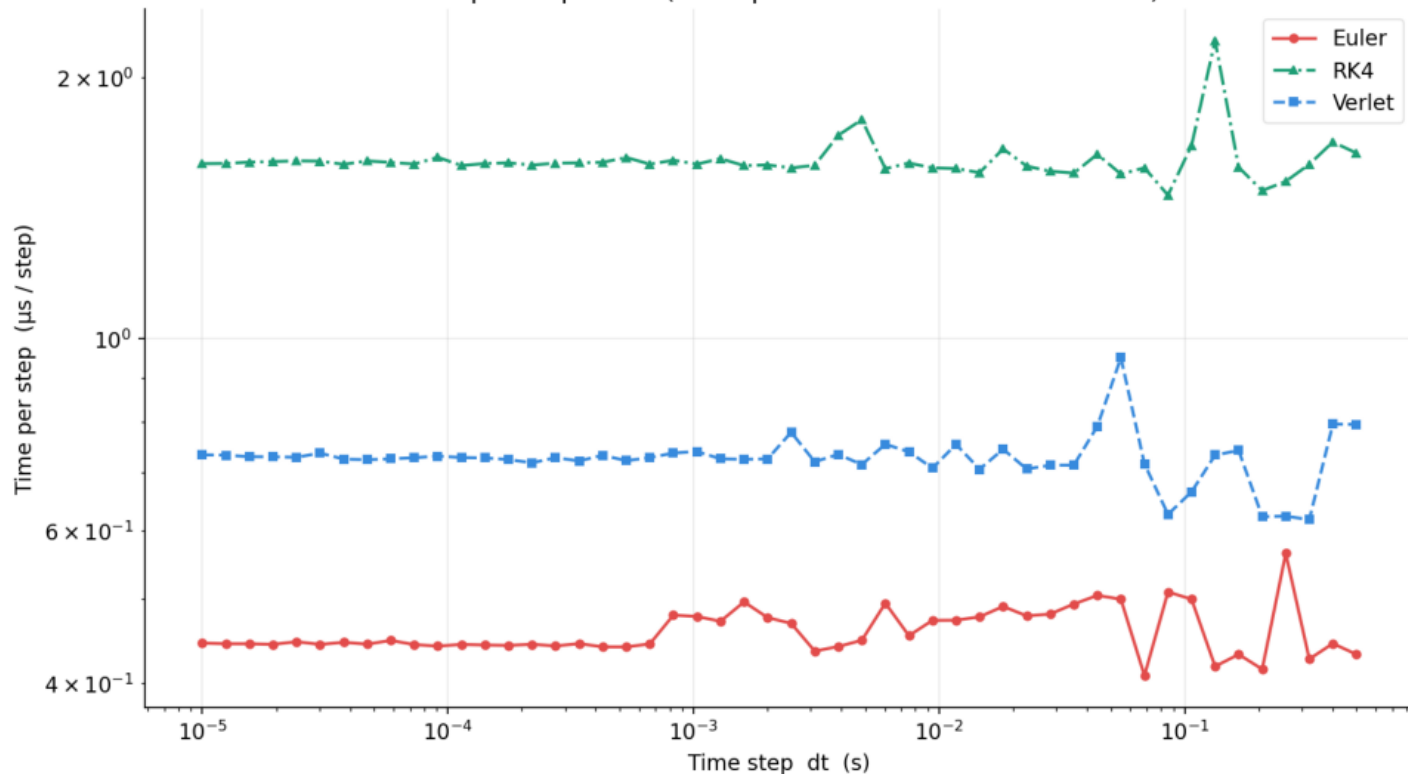
Band interpretation:

Narrow band (small dt) $\rightarrow$
many iterations amortise
OS scheduling noise.

Wide band (large dt) $\rightarrow$
few iterations $\rightarrow$ each run
more sensitive to jitter.

For reliable timing, always
use small dt (many iterations)
or average more runs.

Time per step vs dt (startup overhead visible at coarse dt)



TIME PER STEP

`time_per_step = cpu_mean / N_steps`

Plateau at fine dt = pure integration cost (overhead fully amortised over many steps).

Sustained per-step cost (median of 10 finest-dt runs):

Euler : 0.444 $\mu\text{s}/\text{step}$

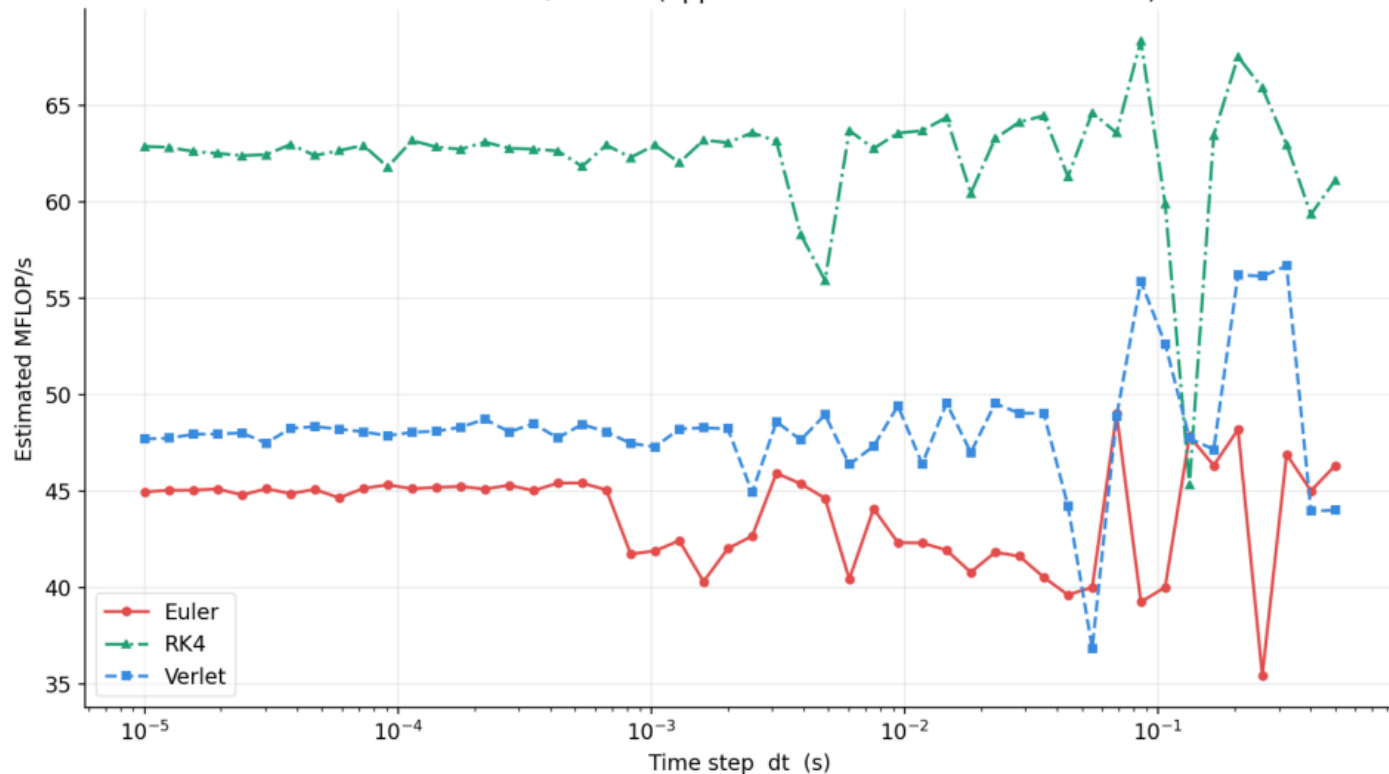
Verlet : 0.730 $\mu\text{s}/\text{step}$

RK4 : 1.597 $\mu\text{s}/\text{step}$

Rising curve at coarse dt:
startup + teardown overhead dominates when only $O(10)$ steps are executed.

Use fine-dt values for real-time per-step budget planning.

Estimated MFLOP/s vs dt (approximate — see notes for caveats)



ESTIMATED MFLOP/s

Sustained MFLOP/s (median of 10 finest-dt runs):

Euler : 45.0 MFLOP/s

Verlet : 48.0 MFLOP/s

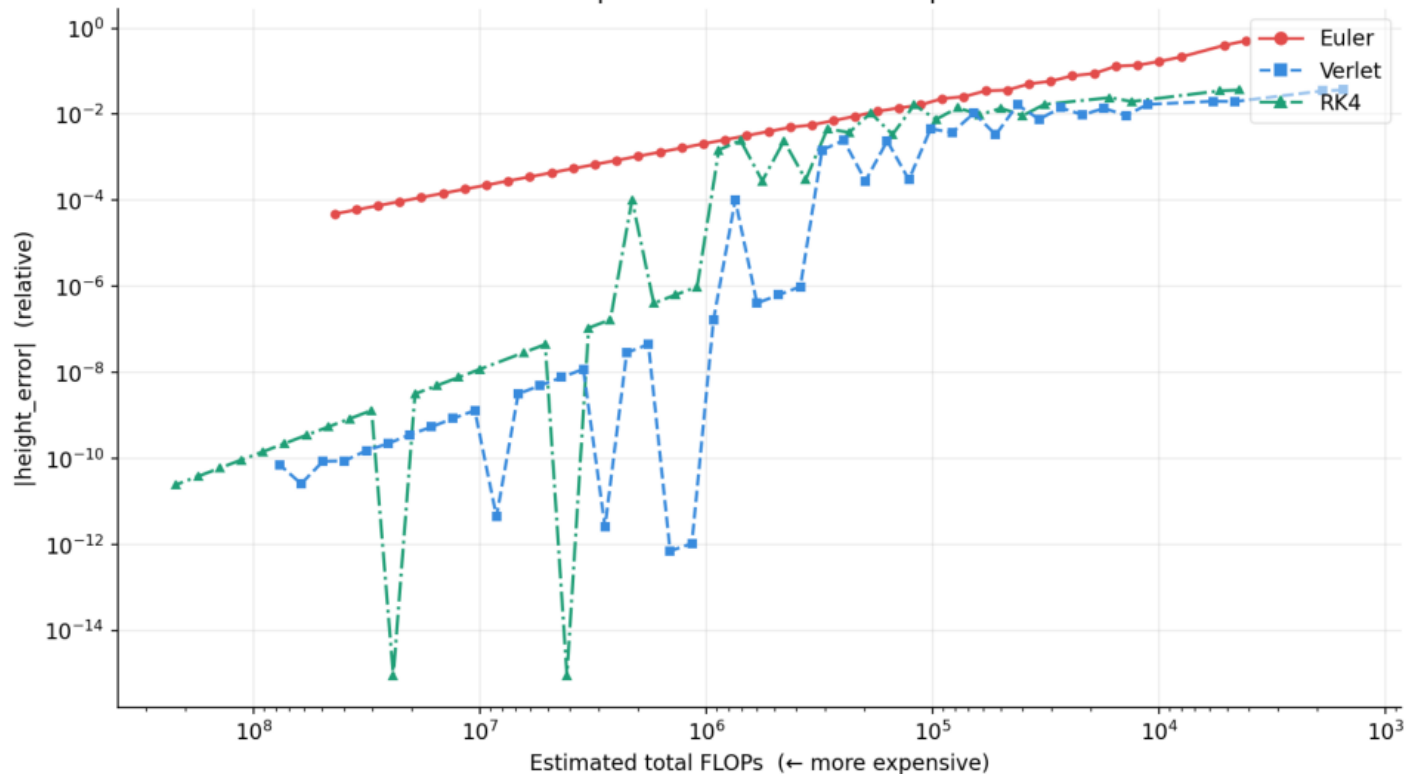
RK4 : 62.6 MFLOP/s

Drop at coarse dt = startup overhead dominates (few iters).

For exact FLOPs: perf / Maqao.

For memory bandwidth: Malt.

Cost vs precision — iso-FLOP comparison



COST vs PRECISION

X = estimated FLOP budget.
Left = more expensive.

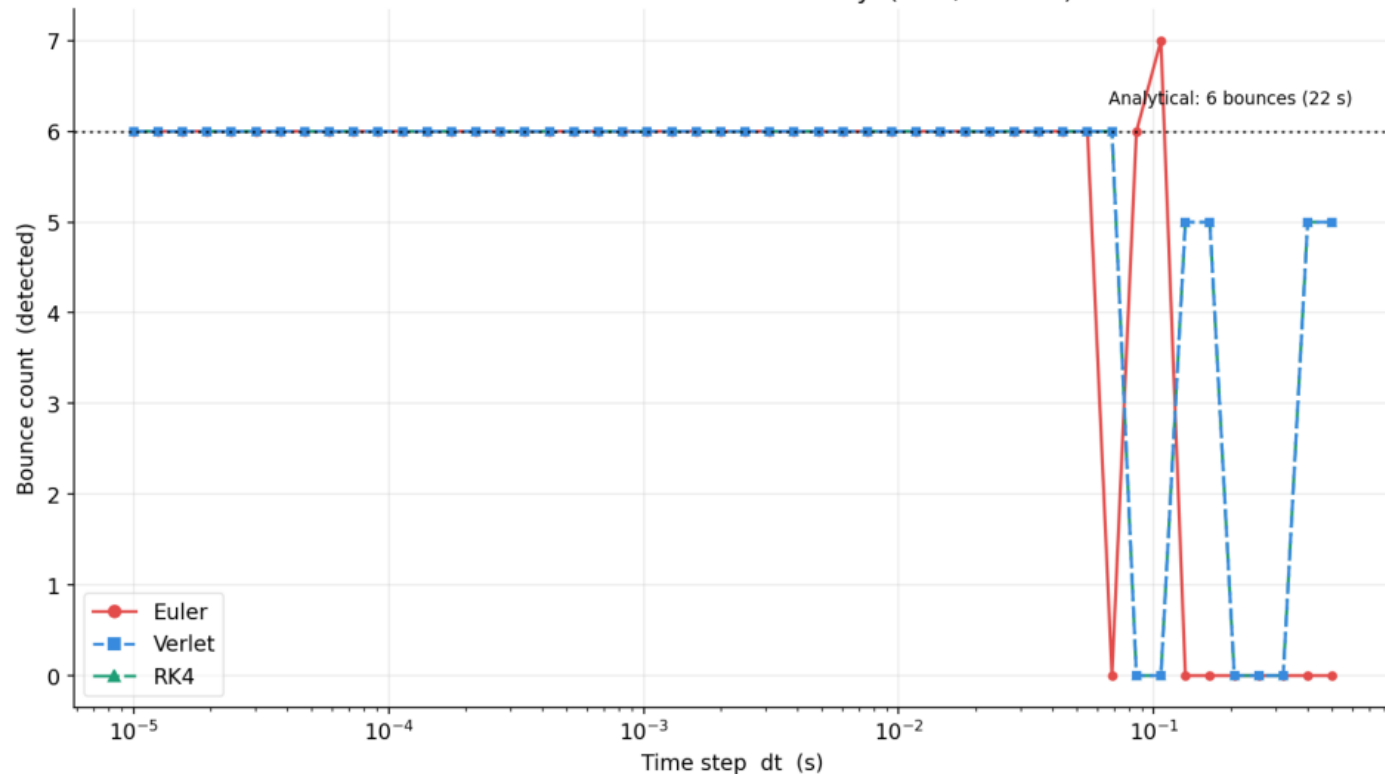
Euler:
 $O(dt^1)$ height error.

Verlet:
 $O(dt^2)$ height error.
Same FLOP budget as Euler
→ Strictly dominates Euler.

RK4:
 $O(dt^2)$ height error.
Slightly lower prefactor
than Verlet at same budget.

Verlet is uniquely optimal:
 $O(dt^2)$ accuracy at $O(dt^1)$
cost – best achievable for
this contact model.

Bounce count vs dt — solver stability (22 s, $e = 1.0$)



BOUNCE COUNT STABILITY

Analytical: 6 bounces in 22 s.
With $e=1$, every bounce returns to $z_0 \rightarrow$ constant flight time.

Tunneling thresholds:

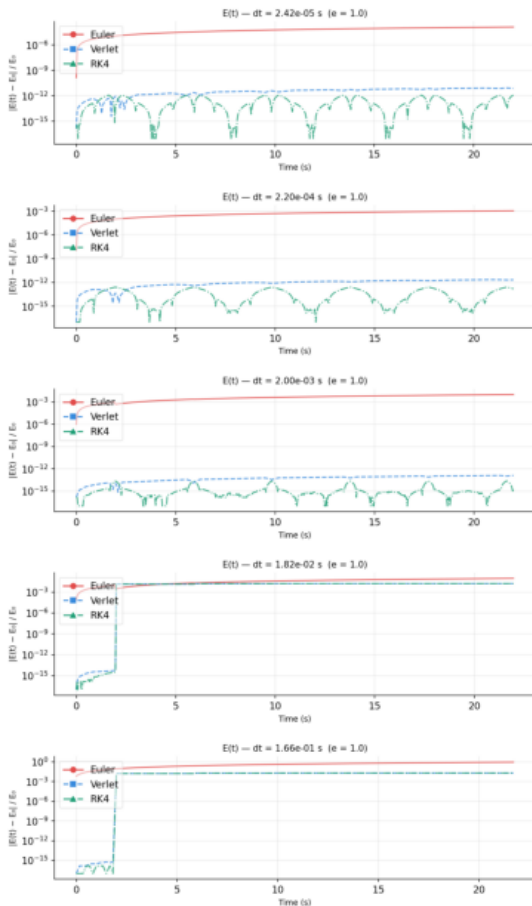
Euler : $dt > 0.069$ s

Verlet/RK4 : $dt > 0.085$ s

Below threshold: all solvers converge to 6 bounces.

Euler transition zone (near tunneling threshold) is wider and less stable than Verlet/RK4.

E(t) time series — energy drift over simulation ($e = 1.0$)



ENERGY DRIFT $E(t)$ ($e = 1.0$)

Each panel: $|E(t) - E_0| / E_0$ over the full simulation.

WD staircase With $e=1.0$:
 Perfectly elastic bounce =
 no physical energy loss.
 All drift is NUMERICAL.

Verlet / RK4:
 Flat at machine precision
 ($\sim 10^{-15}$) for all dt .

Euler:
 Steady upward slope from
 "spiral" dissipation/step.
 Steeper slope at larger dt .

Coarsest dt : ball tunnels.
 Euler: linear drift.
 Verlet/RK4: machine ϵ .